

NICOLÁS ANDRÉS GÓMEZ RAMÍREZ

Electronic Engineering Student | Embedded Systems & PCB Design

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EDUCATION

Pontificia Universidad Javeriana, Bogotá, Colombia
B.S. in Electronic Engineering | GPA: 4.3 / 5.0

Jan 2023 – Expected 2027

TECHNICAL SKILLS

PCB & Hardware Design: Altium Designer (schematic capture, PCB layout; 4+ custom PCBs to date), LTspice, 3D design & printing
Embedded Systems: C, C++, MicroPython; PIC (MPLAB X, MCC), Arduino Uno / Mega 2560, Raspberry Pi (Python, GPIO); SPI, UART, I²C, 1-Wire, PWM, MQTT; finite-state machines (FSM), interrupts, timing analysis
Sensors & Actuation: IR, ADXL345 accelerometer, 12-bit SPI ADC/DAC; SSD1306 OLED, I²C LCD, LED matrices; DC motor control (H-bridge, PWM), servo control
Digital Design: VHDL, Verilog; Intel Quartus, ModelSim testbenches; Cyclone IV E FPGA (DE2-115), VGA timing
Signals: MATLAB; FIR/IIR filter design, FIR on MCU; analog and anti-aliasing filter design; correlation-based detection
Software & Tools: Python, JavaScript/TypeScript, React, Next.js, Firebase, PostgreSQL; Git, Linux, Windows, Vercel
AI: Claude Code, Codex (coding agents); Higgsfield (generative media)
Lab & Instrumentation: Oscilloscope, digital multimeter, bench power supply, function generator, soldering

ACADEMIC EXPERIENCE

Teaching Assistant — Processor-Based Systems Design, Pontificia Universidad Javeriana

Feb 2026 – Present

- Designed and developed custom PCBs in Altium Designer, from schematic capture to layout, for hands-on experimentation with course concepts.
- Resolve office-hours questions on embedded C, finite-state machines, interrupts and timing analysis; support embedded-systems and microprocessor course projects.

Teaching Assistant — Arduino 101, Pontificia Universidad Javeriana

Jan 2025 – Jun 2025

- Guided students through sensors, actuators and peripherals on Arduino; mentored projects from design to implementation and debugged embedded C (memory management, optimization).

PROFESSIONAL EXPERIENCE

Academic Tutor — Programa Tú, Bogotá, Colombia

Sep 2024 – Present

- Provide personalized tutoring in physics, mathematics, critical thinking and language to students at Instituto Alberto Merani (22+ contracted hours).

Freelance Web Developer, Independent businesses

Jul 2026 – Present

- Develop web applications for independent businesses, from requirements definition through implementation and delivery of solutions tailored to their needs.

HARDWARE PROJECTS

PROTEO — Raspberry Pi IoT Node for a Smart Luggage Locker | Python, MQTT, Firebase

2025

- IR sensor published over MQTT (QoS 1) on change, servo latch driven with microsecond pulses (pigpio) and a retained Last-Will status topic; the final prototype syncs a solenoid lock, door state and camera photos with a web dashboard through Firestore.

Real-Time FIR Filtering on a PIC Microcontroller | C, SPI, UART

2025

- Two-channel chain: MAX11666 12-bit ADC → order-12/14 FIR filters in C → MCP4822 12-bit DAC on one shared SPI bus, with a UART command parser and FSM to switch filter, input and routing at runtime.

VOLLEY-PONG — Two-Player Game on an FPGA (DE2-115) | VHDL, team of 2

2024

- 800×600 VGA sync generator verified in ModelSim, fixed-point ball physics with gravity and collisions, boost meter and scoring in pure VHDL; joystick inputs through LM393 comparators.

CORRELACIUM-LEVIOSA — Gesture Recognition by Correlation | MicroPython, I²C, team of 2

2024

- ADXL345 sampled at 100 Hz into 100-sample windows; four spell gestures matched by correlation against recorded templates with per-class thresholds; result on an SSD1306 OLED.

DC Motor Control for a Conveyor Belt | Arduino, C

2025

- Direction and speed from one potentiometer via an L298N H-bridge (PWM), with a dead zone and a friction-aware minimum duty.

DreamSnake — Two-Player Game on Arduino Mega 2560 | C++, team of 3

2023

- Two MAX7219 8×8 LED matrices and a 20×4 I²C LCD, two-player buttons and an FSM-sequenced tune.

LANGUAGES

Spanish: Native | **English:** Upper-Intermediate (B2), Berlitz English Program, 2026